

Bishop's University
CS 596 – Special Topics on Deep Learning

Assignment 3: Multi-label Image Classification

The aim of this final project is to implement a multi-label image classifier on the [PASCAL VOC 2007 dataset](#). You will design and train deep convolutional networks to predict a binary present/absent image-level label for each of the 20 PASCAL classes. This will help you gain experience with PyTorch. You will do the following:

- Part A - Pre-defined Models: Warm-up practice with PyTorch. Run three experiments (listed below) and report your results and observations. *MP3_P1A_Introduction.ipynb* will guide you to start.
 - Train [AlexNet](#) (PyTorch built-in) from scratch.
 - Fine-tune [AlexNet](#) (PyTorch built-in), which is pretrained on [ImageNet](#).
 - Train a simple network (defined in *classifier.py*) from scratch.
- Part B - Self-designed Models: Design your own model architecture to solve this multi-label classification task. *MP3_P1B_Develop_Classifier.ipynb* will guide you to start.
 - No pre-trained model is allowed for Part B.
 - You can use the concepts or ideas from existing models (e.g. VGG, ResNet, DenseNet, etc.), but please don't just copy the existing models. We want to see your own innovation reflected in the model design.
 - You may want to start your design from either a simple network provided in *classifiers.py* or AlexNet.

How to start

You have to rely on the provided starting code. The top-level notebooks (*MP3_P1A_Introduction.ipynb* and *MP3_P1B_Develop_Classifier.ipynb*) will guide you through all the steps of the classifier.

You will be required to use a GPU for this assignment, and hence you will be given

Google Cloud credits shortly. **However, we encourage you to use Colaboratory to get started and for debugging purposes**, as you have limited number of credit hours for this.

Here are two interesting links about Google Colaboratory with the instructions to install Pytorch:

<https://towardsdatascience.com/getting-started-with-google-colab-f2fff97f594c>

<https://colab.research.google.com/notebooks/welcome.ipynb>

By this environment, you will be able to do GPU programming and to run your programs through the cloud computing provided by Google.

Environment Setup (Local)

If you will be completing the assignment on a local machine then you will need a Python environment set up with the appropriate packages.

We suggest that you use [Anaconda](#) to manage python package dependencies. This [guide](#) provides useful information on how to use Conda.

Data Setup (Local)

Once you have downloaded the zip file, create a folder Project and execute the download_data script provided:

```
/download_data.sh
```

Data Setup (For Colaboratory)

If you are using Google Colaboratory for this assignment you will need do some additional setup steps.

You will need to run the download_data.sh script inside colab cells and move all the data to a local directory in colab for fast access.

You will now need to open the assignment 3 ipython notebook file from your Google Drive folder in Colaboratory and run a few setup commands. Make sure to set the GPU

as the hardware accelerator. To do this, on the top bar choose Edit->Notebook Settings->select 'GPU'.

IPython

The final project is given to you in the *MP3_P1A_Introduction.ipynb* and *MP3_P1B_Develop_Classifier.ipynb* file. As mentioned, if you are using Colaboratory, you can open the ipython notebook directly in Colaboratory. If you are using a local machine, ensure that ipython is installed (<https://ipython.org/install.html>). You may then navigate the assignment directory in terminal and start a local ipython server using the jupyter notebook command.

Submission

You will have to submit all the programmed solutions. Please, provide a pdf report following the format of the provided word template that highlights the team members, the contribution of each one of them, and the obtained accuracy for each classifier. The function that computes this accuracy is already provided in the starting code.

Good luck :)